

# UHR-Net: An Uncertainty-Aware Hypergraph Refinement Network for Medical Image Segmentation

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**Abstract**—Accurate lesion segmentation is crucial for clinical diagnosis and treatment planning. However, lesions often resemble surrounding tissues with ill-defined boundaries, and predictions are unstable in ambiguous regions (e.g., boundary and transition areas); meanwhile, small-lesion cues are diluted during multi-scale feature extraction, causing under- or over-segmentation. To address these challenges, we propose an Uncertainty-Aware Hypergraph Refinement Network (UHR-Net). First, we introduce an Uncertainty-Oriented Instance Contrast (UO-IC) pretraining that couples geometry-aware copy-paste augmentation with hard-negative mining from lesion-like backgrounds, enhancing instance discriminability for small and visually ambiguous lesions. Second, we design an Uncertainty-Guided Hypergraph Refinement Block (UGHR Block), which converts a coarse segmentation probability map into an entropy-based uncertainty map to guide hypergraph refinement. We divide hyperedge prototypes into foreground and background groups to decouple higher-order interactions between them, improving refinement in ambiguous regions. Experiments on five public benchmarks demonstrate consistent gains over baselines. Code is available at: <https://github.com/xxxx/UHR-Net>.

**Index Terms**—Medical image segmentation, Uncertainty-Aware Hypergraph Refinement, Uncertainty-Oriented Instance Contrast pretraining

## I. INTRODUCTION

Medical image segmentation is one of the fundamental capabilities in medical image analysis, as it provides critical structured information and quantitative evidence for clinical diagnosis, treatment planning, and therapeutic outcome assessment. In lesion segmentation tasks, automatically and reliably delineating lesion regions and their boundaries is particularly important for subsequent quantitative measurements and decision support.

In recent years, the adoption of deep learning has fostered a variety of automatic segmentation approaches, substantially improving the accuracy and efficiency of medical image segmentation. Nevertheless, in real-world clinical images, lesion segmentation accuracy is still hindered by two types of uncertainty. On the one hand, for instance-level discrimination and localization, small lesions occupy only a tiny fraction of pixels, and their discriminative cues are easily diluted during multi-scale feature extraction, leading to missed lesions or over-segmentation. On the other hand, for pixel-level fine-grained prediction, lesions often resemble surrounding tissues and exhibit gradually transitioning boundaries. As a result, ambiguous regions lack clear separation cues and stable discriminative cues, making models more prone to confusion in these areas.

To tackle these challenges, existing studies have explored multiple directions. Convolutional neural network (CNN)-based methods (e.g., HarDNet-CPS [1], LDNet [2]) typically enhance multi-scale fusion and detail recovery within U-shaped encoder-decoder frameworks, improving adaptability to complex morphologies and low-contrast lesions by refining multi-scale aggregation or introducing more flexible prediction heads. However, in many CNN segmentation approaches, uncertainty is often leveraged indirectly (e.g., via loss re-weighting, regularization terms, or hard-sample mining), rather than being explicitly transformed into a guiding signal for subsequent structured refinement or inference. To alleviate the limitations of local modeling, Transformer-based methods (e.g., H2Former [3]) use self-attention to capture long-range dependencies and strengthen global interactions, while graph neural network (GNN) methods (e.g., ViG-UNet [4]) construct sparse graph structures and employ message passing for more efficient global dependency modeling. Nevertheless, the core interactions in both families are still dominated by token-token or node-node pairwise relationships, which makes it difficult to explicitly model many-to-many higher-order consistency jointly determined by multiple regions.

In this paper, we propose UHR-Net, an uncertainty-aware hypergraph refinement network for lesion segmentation. To address cue dilution in small lesions and spurious foreground responses induced by lesion-like backgrounds, we introduce Uncertainty-Oriented Instance Contrast (UO-IC) to pretrain the backbone. UO-IC uses geometry-aware copy-paste to generate images containing an original lesion instance and its downscaled replica, forming instance-level positive pairs, while mining lesion-like backgrounds as hard negatives. Contrastive learning among the original instance, the scaled instance, and the background strengthens instance-level discriminative representations. To improve prediction consistency in ambiguous regions, we further design an Uncertainty-Guided Hypergraph Refinement Block (UGHR Block). It derives an entropy-based uncertainty map  $U$  from the coarse segmentation probability map  $\hat{M}$  to modulate node-hyperedge participation logits, strengthening context aggregation in ambiguous regions and enforcing many-to-many higher-order consistency. Moreover, foreground- and background-conditioned hyperedge prototypes decouple higher-order aggregation pathways, reducing boundary interference. Extensive experiments on ISIC-2016, ISIC-2017, GlaS, Kvasir-SEG, and Kvasir-Sessile demonstrate consistent gains and competitive performance.

Overall, the main contributions of this paper are as follows:

- We propose UHR-Net, an uncertainty-aware hypergraph refinement framework that couples UO-IC instance-discriminative pretraining with structured refinement for ambiguous regions.
- We introduce UO-IC, combining geometry-aware copy-paste and lesion-like hard-negative mining to strengthen instance-level discriminative representations.
- We develop the UGHR Block, which derives an entropy-based uncertainty map to modulate node-hyperedge participation logits and employs foreground/background-conditioned hyperedge prototypes to decouple higher-order interactions; experiments on five public benchmarks and ablations verify consistent gains.

## II. METHODOLOGY

### A. Overview

To alleviate the dilution of small-lesion representations, spurious foreground responses induced by lesion-like backgrounds, and unstable predictions in ambiguous regions in medical lesion segmentation, we propose UHR-Net. The core idea is to couple instance-level discriminative representation enhancement with structured refinement for ambiguous regions: during pretraining, we guide the backbone to learn finer-grained and more discriminative cues; during end-to-end training, we introduce a coarse segmentation probability map and an uncertainty-guided refinement mechanism implemented by multi-scale UGHR Blocks, so that refinement focuses more on ambiguous regions, thereby improving prediction stability.

As shown in Fig. 1, given an input image  $I$ , UHR-Net outputs the final foreground probability map  $\hat{Y}$ , with the corresponding ground truth denoted as  $Y$ . The training pipeline consists of two parts: UO-IC pretraining and end-to-end training. In UO-IC, geometry-aware copy-paste constructs positive pairs and mines lesion-like hard negatives under an InfoNCE objective. In end-to-end training, a lightweight guidance head produces a coarse probability map  $\hat{M}$ , from which an entropy-based uncertainty map  $U$  is derived to guide multi-scale UGHR Blocks. In each UGHR Block, the UGHR branch builds foreground- and background-conditioned hyperedge prototypes and modulates node-hyperedge participation logits with  $U$ , while a parallel Dil.Conv branch provides local enhancement. Finally, the decoder aggregates refined features and outputs  $\hat{Y}$ . Details are provided in Section II-C.

### B. Uncertainty-Oriented Instance Contrast

Existing pretraining strategies that rely solely on standard pixel-wise supervision, or adopt contrastive objectives with generic data augmentations, often struggle to specifically alleviate two key sources of segmentation errors: (i) small-lesion cues are easily diluted during multi-scale feature extraction, and (ii) lesion-like backgrounds tend to trigger spurious foreground responses. To this end, we propose UO-IC, an instance-level contrastive pretraining strategy tailored for lesion segmentation. It combines geometrically constrained

copy-paste augmentation with lesion-like hard-negative mining, and employs an InfoNCE objective to guide the backbone to learn more stable and discriminative representations.

To perform copy-paste under geometric constraints, we extract the lesion instance mask and its bounding box from the ground-truth mask  $Y$  for cropping and scaling. Meanwhile, we compute a safety distance map  $D_{\text{dist}}$  by applying a Euclidean distance transform on the background region  $(1 - Y)$ , where  $D_{\text{dist}}(x)$  denotes the distance from a background pixel  $x$  to its nearest foreground pixel. During training, we randomly select a lesion instance in the current image as *Lesion A* and generate a scaled replica *Lesion B*. To avoid overlap between the pasted instance and existing foreground regions, we define a safety radius  $r_s$  for Lesion B (as the radius of the circumscribed circle of its bounding box), and select a paste center  $p$  on  $D_{\text{dist}}$  such that  $D_{\text{dist}}(p) \geq r_s$ . We then paste Lesion B at a location satisfying this constraint to obtain the augmented image  $I^{cp}$ , and merge its mask into  $Y$  to produce the supervision mask  $Y^{cp}$ . Meanwhile, we retain two single-instance masks,  $M_A$  (the original instance A) and  $M_B$  (the scaled-and-pasted replica B), for subsequent representation extraction and contrastive learning.

During the forward pass, we feed  $I^{cp}$  into the backbone to obtain a feature map  $\tilde{F}$ . We then perform masked average pooling over  $\tilde{F}$  within the regions specified by  $M_A$  and  $M_B$ , yielding the instance representations  $\mathbf{z}_A$  (the original instance A) and  $\mathbf{z}_B$  (its scaled-and-pasted replica B), which form a positive pair for the InfoNCE objective.

To construct hard negatives, we focus on lesion-like background locations in the background region  $(1 - Y^{cp})$  where the predicted foreground probability map  $\hat{Y}^{cp}$  is high. Specifically, we perform weighted masked average pooling (WMAP) on features using  $\hat{Y}^{cp}$  as the weighting term, obtaining the lesion-like background representation

$$\mathbf{z}_{bg} = \text{WMAP}\left(\tilde{F}, (1 - Y^{cp}) \odot \hat{Y}^{cp}\right). \quad (1)$$

During training, we compute one lesion-like background representation per sample and use the in-batch set  $\mathbf{z}_{bg}^{(k)}$  as hard negatives to contrast against  $\mathbf{z}_A$ .

Based on the above representations, we adopt the InfoNCE loss [5] for instance-level contrastive optimization, where  $\text{sim}(\cdot, \cdot)$  denotes cosine similarity. Taking  $\mathbf{z}_A$  as the anchor,  $\mathbf{z}_B$  as the positive sample, and  $\{\mathbf{z}_{bg}^{(k)}\}_{k=1}^{N_{\text{neg}}}$  as the negative set (where  $N_{\text{neg}}$  is the number of negatives in the mini-batch), the InfoNCE objective is defined as

$$\begin{aligned} s^+ &= \text{sim}(\mathbf{z}_A, \mathbf{z}_B) / \tau, \\ s_k^- &= \text{sim}(\mathbf{z}_A, \mathbf{z}_{bg}^{(k)}) / \tau. \end{aligned} \quad (2)$$

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(s^+)}{\exp(s^+) + \sum_{k=1}^{N_{\text{neg}}} \exp(s_k^-)}. \quad (3)$$

This objective pulls  $\mathbf{z}_A$  closer to its scaled-and-pasted replica  $\mathbf{z}_B$  while pushing  $\mathbf{z}_A$  away from the lesion-like background representations. Here,  $\tau$  is the temperature parameter that scales the contrastive logits.

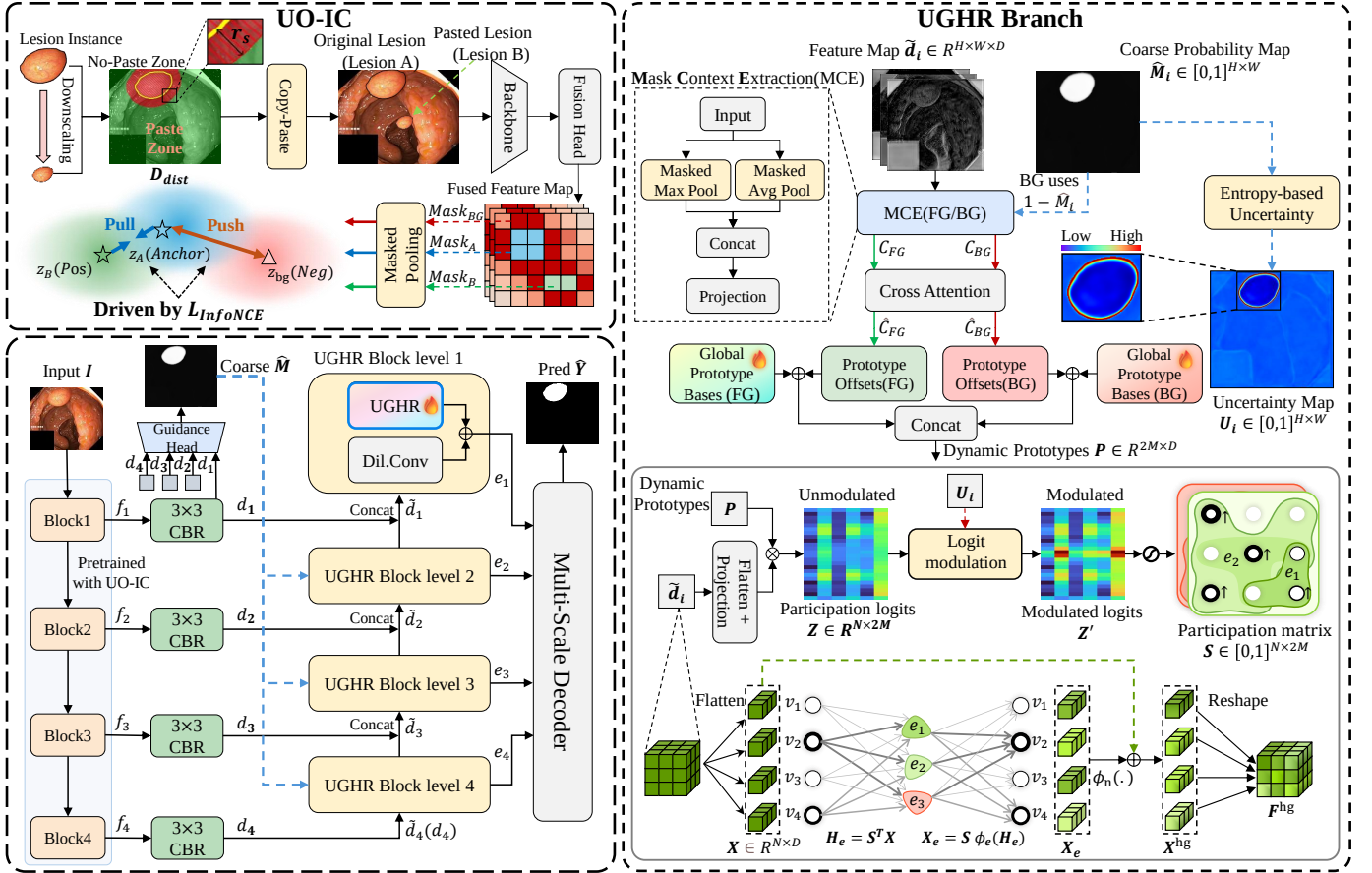


Fig. 1. Overall framework of UHR-Net, consisting of UO-IC pretraining and a multi-scale decoder equipped with UGHR Blocks for uncertainty-guided hypergraph refinement.

Moreover, to avoid degrading pixel-wise localization by relying solely on instance-level contrastive constraints, we also impose segmentation supervision during pretraining; the complete objective is provided in Section II-D.

### C. Uncertainty-Guided Hypergraph Refinement Block

Along the refinement path, the UGHR Block at the  $i$ -th scale takes a fused input feature  $\tilde{d}_i$ , which is obtained by combining the channel-aligned feature at the current scale with cross-scale information from the deeper refined output. Meanwhile, the coarse segmentation probability map  $\hat{M}$  is resized to this scale to obtain  $\hat{M}_i$ . Given  $(\tilde{d}_i, \hat{M}_i)$ , the UGHR Block outputs the refined feature  $e_i$ . Since ambiguous regions typically have high uncertainty and prediction fluctuations, the UGHR branch first computes a pixel-wise uncertainty map  $U_i$  from  $\hat{M}_i$ , and uses  $U_i$  to modulate the node-hyperedge participation logits (before normalization) (the normalized participation weights are computed with respect to the foreground- and background-conditioned hyperedge groups; see Section II-C1). This modulation amplifies the participation logits at highly uncertain locations, encouraging larger normalized participation weights after normalization, so that subsequent hypergraph message passing can more effectively aggregate contextual information from these regions, improving prediction consistency in ambiguous regions. Next, we introduce uncertainty-guided hyper-

graph construction (Section II-C1) and hypergraph message passing and feature updating (Section II-C2) in turn.

1) *Uncertainty-Guided Hypergraph Construction:* We compute a pixel-wise uncertainty map  $U_i$  from the coarse segmentation probability map  $\hat{M}_i$ :

$$U_i = -\frac{\hat{M}_i \log(\hat{M}_i + \varepsilon) + (1 - \hat{M}_i) \log(1 - \hat{M}_i + \varepsilon)}{\log 2}, \quad (4)$$

where  $\varepsilon$  is a numerical stability constant, and  $\log 2$  normalizes the binary entropy to  $[0, 1]$ . When  $\hat{M}_i$  is close to 0.5,  $U_i$  attains larger values, which typically correspond to ambiguous regions. In what follows,  $U_i$  is used to modulate the node-hyperedge participation logits, yielding normalized participation weights.

Next, we treat each pixel location in  $\tilde{d}_i$  as a node and introduce two groups of hyperedge prototypes, i.e., foreground and background, so that hyperedges can selectively aggregate node information over the entire image. Prototype generation is conditioned on the coarse segmentation prior: we use  $\hat{M}_i$  and  $1 - \hat{M}_i$  as soft weights for the foreground and background, respectively, and adopt a lightweight Mask Context Extraction (MCE) module to perform mask-guided pooling on  $\tilde{d}_i$ , producing a foreground context vector  $c_{FG}$  and a background context vector  $c_{BG}$ . Considering that coarse predictions in ambiguous regions can be mixed, we further apply bidirectional cross-

attention between the two contexts to obtain  $\tilde{c}_{FG}$  and  $\tilde{c}_{BG}$ , which are used to generate sample-wise offsets and thus construct sample-wise dynamic prototypes.

Specifically, we maintain two groups of learnable global prototype bases  $P_{FG}^g$  and  $P_{BG}^g$ , where each group contains  $M$  hyperedge prototypes and  $P_{FG}^g, P_{BG}^g \in \mathbb{R}^{M \times D}$ . Here,  $D$  denotes the node embedding dimension (i.e., the channel dimension of  $\tilde{d}_i$ ). Given the interacted contexts  $\tilde{c}_{FG}$  and  $\tilde{c}_{BG}$ , we generate sample-wise prototype offsets  $\Delta P_{FG}, \Delta P_{BG} \in \mathbb{R}^{M \times D}$ , and add them to the corresponding bases to obtain dynamic foreground and background prototypes:

$$P_{FG} = P_{FG}^g + \Delta P_{FG}, \quad P_{BG} = P_{BG}^g + \Delta P_{BG}. \quad (5)$$

We then concatenate the two prototype sets to obtain the sample-specific dynamic prototypes

$$P = \text{Concat}(P_{FG}, P_{BG}) = [p_1, \dots, p_{2M}] \in \mathbb{R}^{2M \times D}. \quad (6)$$

where the first  $M$  prototypes belong to the foreground group and the remaining  $M$  prototypes belong to the background group. The dynamic prototypes  $P$  are used to compute node-hyperedge participation logits.

To explicitly inject uncertainty into node-hyperedge participation assignment, let  $u_n \in [0, 1]$  denote the  $n$ -th element of the flattened uncertainty map  $U_i$ . After obtaining  $P$ , we flatten  $\tilde{d}_i$  along the spatial dimensions into a node feature matrix  $X = [x_1, \dots, x_N]^\top \in \mathbb{R}^{N \times D}$ , and compute unnormalized node-hyperedge logits via scaled dot-product:

$$z_{n,m} = \frac{x_n^\top p_m}{\sqrt{D}}, \quad n = 1, \dots, N, \quad m = 1, \dots, 2M. \quad (7)$$

We then modulate  $z_{n,m}$  pointwise by  $u_n$  and normalize over the node dimension for each hyperedge to obtain the participation matrix  $S \in \mathbb{R}^{N \times 2M}$ :

$$z'_{n,m} = z_{n,m} \cdot 2^{\beta u_n}, \quad S_{n,m} = \frac{\exp(z'_{n,m})}{\sum_{n'=1}^N \exp(z'_{n',m})}, \quad m = 1, \dots, 2M, \quad (8)$$

where  $S_{n,m}$  denotes the normalized participation weight of node  $n$  in hyperedge  $m$ , and  $\beta$  is the modulation strength. This uncertainty-aware logit modulation increases the normalized participation weights of high-uncertainty nodes in  $S$ , thereby emphasizing their contributions during subsequent hypergraph message passing and feature updates.

#### 2) Hypergraph Message Passing and Feature Updating:

After obtaining the participation matrix  $S$  and the node features  $X$ , we perform a two-step message passing scheme on the hypergraph to update node representations. First, nodes are aggregated into hyperedge representations using the participation matrix  $S$ ,  $H_e = S^\top X$ . Then,  $H_e$  is projected by an edge mapping  $\phi_e(\cdot)$  and propagated back to nodes using  $S$ , yielding the returned messages  $X_e$ . The messages are further transformed by a node mapping  $\phi_n(\cdot)$  and added to the input node features  $X$  via a residual connection to obtain the updated node features  $X^{hg}$ :

$$X_e = S \phi_e(H_e), \quad X^{hg} = X + \phi_n(X_e). \quad (9)$$

Reshaping  $X^{hg}$  back to the spatial layout gives the output of the UGHR branch, denoted as  $F^{hg}$ . Meanwhile, the Dil.Conv branch produces a locally enhanced feature map  $F^{conv}$ . We fuse the two branch outputs in a pixel-wise manner to obtain the refined feature  $e_i$ .

#### D. Loss Function

Training consists of two phases: UO-IC pretraining and end-to-end training. In both phases, we use a combined Dice and BCE loss for segmentation supervision, denoted as  $\mathcal{L}_{seg}(\hat{Y}, Y)$ .

During pretraining, we incorporate the instance-level contrastive term in Section II-B on top of the segmentation supervision. The pretraining objective is

$$\mathcal{L}_{pre} = \mathcal{L}_{seg}(\hat{Y}^{cp}, Y^{cp}) + \lambda_{ic} \mathcal{L}_{InfoNCE}. \quad (10)$$

During end-to-end training, in addition to producing the final prediction  $\hat{Y}$ , the guidance head also outputs a coarse segmentation probability map  $\hat{M}$ . To keep the coarse guidance stable during training and encourage it to learn a coarse prior aligned with the final segmentation objective, we apply an auxiliary segmentation supervision to the upsampled coarse map  $\hat{M}_\uparrow$  (upsampled to the output resolution). The overall training objective is

$$\mathcal{L}_{all} = \mathcal{L}_{seg}(\hat{Y}, Y) + \lambda_{aux} \mathcal{L}_{seg}(\hat{M}_\uparrow, Y). \quad (11)$$

where  $\lambda_{ic}$  and  $\lambda_{aux}$  denote the weights of the contrastive term and the auxiliary supervision, respectively.

### III. EXPERIMENTS

#### A. Experimental Setup

We evaluate UHR-Net on five public datasets: Kvasir-SEG [20], Kvasir-Sessile [21], GlaS [22], ISIC-2016 [23], and ISIC-2017 [24]. We report mIoU and mDSC on all datasets, and additionally Recall and Precision on Kvasir-SEG, Kvasir-Sessile, and GlaS. All experiments are implemented in PyTorch, and dataset splits and additional statistics are provided in the supplementary material. We use Adam with a learning rate of  $1 \times 10^{-4}$ : (i) UO-IC pretraining uses a batch size of 16 with  $\tau = 0.10$  and  $\lambda_{ic} = 1.0$ ; (ii) end-to-end training uses a batch size of 24 with  $\lambda_{aux} = 0.1$ . For the UGHR branch, we set  $M = 8$  prototypes per group (total  $2M$ ) and  $\beta = 1.0$ .

#### B. Comparison with Other Methods

Tables I and II compare UHR-Net with representative methods on five datasets. Overall, UHR-Net achieves the best mIoU on all five datasets and consistently leads in mDSC, Recall, and Precision.

#### C. Ablation Study

Unless otherwise specified, the experimental results reported in Tables III to V are obtained on the Kvasir-SEG dataset. To validate the effectiveness of the key designs in our framework, we conduct ablation studies on UO-IC and the UGHR Block, with results summarized in Table III. Specifically, Experiment

TABLE I  
COMPARISON WITH OTHER METHODS ON THE KVASIR-SESSILE, KVASIR-SEG AND GLAS DATASETS. **BOLD** INDICATES THE BEST RESULT AND UNDERLINED INDICATES THE SECOND BEST.

| Methods        | Kvasir-Sessile |             |             |             | Kvasir-SEG  |             |             |             | GlaS        |             |             |             |
|----------------|----------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                | mIoU           | mDSC        | Rec.        | Prec.       | mIoU        | mDSC        | Rec.        | Prec.       | mIoU        | mDSC        | Rec.        | Prec.       |
| U-Net [6]      | 23.1           | 33.8        | 45.1        | 46.6        | 65.5        | 75.8        | 83.6        | 77.6        | 75.8        | 85.5        | 90.3        | 82.8        |
| U-Net++ [7]    | 38.4           | 50.2        | 62.5        | 51.8        | 67.9        | 77.2        | 86.5        | 77.6        | 77.6        | 86.9        | 89.6        | 85.5        |
| PraNet [8]     | 66.7           | 77.4        | 80.7        | 82.4        | 83.0        | 89.4        | 90.6        | 91.3        | 71.8        | 83.0        | 90.9        | 78.0        |
| TGANet [9]     | 74.4           | 82.0        | 79.3        | 85.9        | 83.3        | 89.8        | 91.3        | 91.2        | 71.8        | 84.7        | 86.9        | 80.2        |
| DCSAU-Net [10] | 72.6           | 81.1        | 65.6        | 62.9        | 83.5        | 88.9        | 89.5        | 89.5        | 77.6        | 86.5        | 93.0        | 82.5        |
| CASF-Net [11]  | 60.5           | 72.4        | 78.0        | 74.8        | 81.7        | 88.7        | 89.2        | 88.2        | 78.4        | 87.2        | 91.3        | 85.9        |
| DTAN [12]      | 76.4           | 84.2        | 84.2        | 85.9        | 84.1        | 90.4        | 91.6        | <b>92.0</b> | 78.5        | 87.9        | 85.8        | 90.2        |
| CMUNeXt [13]   | 75.3           | 85.2        | 89.6        | 82.9        | 81.1        | 89.0        | 89.3        | 89.9        | 81.2        | 89.1        | 89.2        | 90.1        |
| ESPNet [14]    | <u>82.4</u>    | <u>90.4</u> | 89.3        | <b>91.4</b> | 83.6        | <b>91.1</b> | 90.9        | 91.2        | 85.0        | <u>91.9</u> | <u>94.3</u> | 89.6        |
| ConDSeg [15]   | 81.2           | 89.1        | <u>90.1</u> | <u>90.0</u> | <u>84.6</u> | 90.5        | <u>92.3</u> | <u>91.7</u> | <u>85.1</u> | 91.6        | 93.5        | <u>90.5</u> |
| <b>Ours</b>    | <b>83.5</b>    | <b>90.6</b> | <b>91.4</b> | <b>91.4</b> | <b>85.0</b> | <u>90.8</u> | <b>92.7</b> | 91.2        | <b>87.0</b> | <b>92.7</b> | <b>94.7</b> | <b>91.5</b> |

TABLE II  
COMPARISON WITH OTHER METHODS ON THE ISIC-2016 AND ISIC-2017 DATASETS. SAME NOTATION AS TABLE I.

| Methods        | ISIC-2016   |             | ISIC-2017   |             |
|----------------|-------------|-------------|-------------|-------------|
|                | mIoU        | mDSC        | mIoU        | mDSC        |
| U-Net [6]      | 83.6        | 90.3        | 73.7        | 82.8        |
| CENet [16]     | 84.6        | 90.9        | 76.4        | 84.8        |
| FAT-Net [17]   | 85.3        | 91.6        | 76.5        | 85.0        |
| DCSAU-Net [10] | 85.3        | 91.4        | 76.1        | 85.0        |
| BGDiffSeg [18] | 85.5        | 92.2        | 79.7        | 88.7        |
| MSCB-UNet [19] | 84.2        | 91.4        | <u>81.2</u> | <b>89.6</b> |
| ConDSeg [15]   | <u>86.8</u> | <u>92.5</u> | 80.9        | 88.3        |
| <b>Ours</b>    | <b>87.6</b> | <b>92.9</b> | <b>81.8</b> | <u>89.2</u> |

TABLE III  
ABLATION STUDY ON KEY COMPONENTS. CHECKMARKS INDICATE ENABLED COMPONENTS. ABBREVIATIONS: BASE HR = BASE HYPERGRAPH REFINER; UNC. GUIDE. = UNCERTAINTY GUIDANCE; FG/BG HG = FG/BG HYPEREDGE GROUPS.

|     | UGHR Branch           |         |             |          | Metric      |             |
|-----|-----------------------|---------|-------------|----------|-------------|-------------|
|     | Pre-training<br>UO-IC | Base HR | Unc. Guide. | FG/BG HG | mIoU        | mDSC        |
| (1) | -                     | -       | -           | -        | 81.2        | 88.8        |
| (2) | ✓                     | -       | -           | -        | 82.2        | 89.0        |
| (3) | -                     | ✓       | -           | -        | 83.0        | 89.5        |
| (4) | -                     | ✓       | -           | ✓        | 83.5        | 89.8        |
| (5) | -                     | ✓       | ✓           | -        | 83.9        | 90.1        |
| (6) | -                     | ✓       | ✓           | ✓        | 84.4        | 90.3        |
| (7) | ✓                     | ✓       | ✓           | ✓        | <b>85.0</b> | <b>90.8</b> |

(Exp.) 1 corresponds to the baseline model, and Exp. 2 enables UO-IC. Exps. 3 to 6 progressively evaluate three designs in the UGHR branch, namely Base HR, Unc. Guide., and FG/BG HG, while Exp. 7 is the full model.

The switches and their implementation correspondences are as follows. When Base HR is turned off, we replace the UGHR branch in the UGHR Block with an equivalent standard convolutional module to remove hypergraph-based higher-order aggregation. When Unc. Guide. is turned off, hypergraph message passing and feature updating are retained, but the entropy-based uncertainty map  $U$  is no longer used to modulate the node-hyperedge participation logits. When FG/BG HG is turned off, the total number of hyperedge prototypes is kept

TABLE IV  
ABLATION ON THE NUMBER OF HYPEREDGE PROTOTYPES PER GROUP ( $M$ ).

| $M$ | mIoU         | mDSC         | Recall       | Precision    |
|-----|--------------|--------------|--------------|--------------|
| 2   | 83.46        | 89.48        | 94.48        | 88.21        |
| 4   | 84.02        | 90.28        | <b>94.62</b> | 88.57        |
| 8   | <b>85.00</b> | <b>90.80</b> | 92.70        | 91.20        |
| 16  | 84.65        | 89.97        | 91.81        | <b>91.42</b> |
| 32  | 84.54        | 89.93        | 94.09        | 89.43        |

TABLE V  
ABLATION ON LOGIT MODULATION STRENGTH ( $\beta$ ).

| $\beta$ | mIoU         | mDSC         | Recall       | Precision    |
|---------|--------------|--------------|--------------|--------------|
| 0.1     | 83.66        | 89.90        | <b>95.97</b> | 86.86        |
| 0.5     | 84.19        | 90.18        | 93.16        | 89.53        |
| 1.0     | <b>85.00</b> | <b>90.80</b> | 92.70        | 91.20        |
| 1.5     | 84.28        | 89.76        | 91.73        | <b>91.86</b> |
| 2.0     | 84.17        | 89.91        | 94.18        | 89.16        |

fixed at  $2M$ , but we no longer split prototypes into foreground and background groups with conditional generation; instead, we generate a shared set of  $2M$  dynamic hyperedge prototypes based on global context.

Table III shows that introducing UO-IC alone (Exp. 2) already improves the baseline, indicating that the proposed pretraining strengthens instance-level discriminability even without hypergraph refinement. Enabling Base HR (Exp. 3) brings a clear gain over Exp. 1, validating that hypergraph-based higher-order aggregation is beneficial for refining predictions. On top of Base HR, FG/BG HG (Exp. 4) and Unc. Guide. (Exp. 5) each further improves performance: FG/BG HG reduces foreground-background interference by decoupling their prototype groups, while Unc. Guide. injects the uncertainty map  $U$  into participation logits to steer message passing toward ambiguous regions. When both designs are enabled (Exp. 6), the gains accumulate, and the full model (Exp. 7) achieves the best results, suggesting that UO-IC and the UGHR branch are complementary.

**Number of hyperedge prototypes  $M$ .** We evaluate the effect of the per-group hyperedge prototype number  $M$  in the UGHR branch on performance (Table IV). Here,  $M$  is the per-group

number (total  $2M$  prototypes). When  $M$  is too small (e.g., 2 or 4), the prototype capacity is insufficient and the refinement gain is limited; when  $M$  is too large (e.g., 16 or 32), it tends to introduce redundancy and additional noise. Overall,  $M = 8$  achieves the best performance, and thus we use  $M = 8$  as the default setting in subsequent experiments.

**Logit modulation strength  $\beta$ .** In Table V, we evaluate the effect of  $\beta$  in (8) (i.e., the modulation factor  $2^{\beta u_n}$ ) on the strength of uncertainty modulation. When  $\beta$  is small (e.g., 0.1), the modulation is insufficient, making it difficult to effectively enhance the focus on highly ambiguous regions. When  $\beta$  is too large (e.g., 1.5 or 2.0), over-modulation may occur and undermine training stability. Overall,  $\beta = 1.0$  achieves the best performance; therefore, we set  $\beta = 1.0$  as the default in subsequent experiments.

#### IV. CONCLUSION

In this paper, we present UHR-Net, an uncertainty-aware hypergraph refinement network for medical lesion segmentation that tackles cue dilution in small lesions and prediction instability in ambiguous regions. We introduce UO-IC pre-training to enhance instance discriminability via geometry-aware copy-paste and lesion-like hard-negative mining. We further develop UGHR Blocks that (i) derive an entropy-based uncertainty map to modulate node-hyperedge participation logits; and (ii) employ foreground- and background-conditioned hyperedge prototypes to decouple higher-order interactions. Extensive experiments on five public benchmarks demonstrate consistent gains over representative methods.

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